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Daimler Center for Automotive IT Innovations

DCAITI-Project: Big Data Storage Solution for Mobility Data

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Overview

1. Context
2. Drawbacks
3. Goals
4. Plan
   4.1. Research
   4.2. Installation and Benchmarking
5. Roadmap
6. References

Apache Accumulo [1]

- NoSQL.
- Distributed.
- Sorted key/value store.
- Wide Column.
- Google's BigTable paper. [2]
- Apache Hadoop's HDFS.
- Apache ZooKeeper
- Cell based security.
- CP system (CAP).
- Linear scaling.
- Automatic load balancing.
- High availability.
- Geospatial data support. [3]

[4]

[5]

[6]

Drawbacks

- Development pace.
  - Apache Incubation: 2011 [7]

Apache Accumulo

Developer(s)	Apache Software Foundation
Stable release	2.0.1 (December 24, 2020; 2 years ago[1]) [±]
Repository	Accumulo Repository
Written in	Java
Operating system	Cross-platform
License	Apache License 2.0
Website	accumulo.apache.org

[8]

Apache Accumulo

accumulo.apache.org

big-data accumulo hacktoberfest

Readme

Apache-2.0 license

Code of conduct

Security policy

973 stars

90 watching

424 forks

Report repository

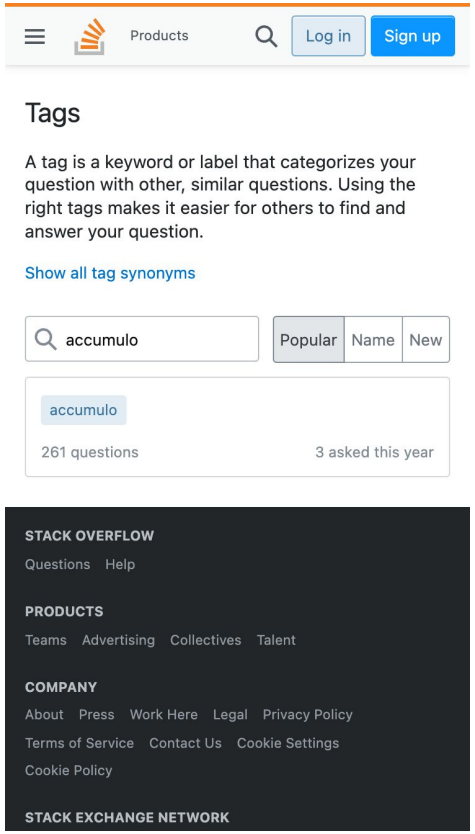
Releases

41 tags

[9]

Drawbacks

- Low popularity.
  - 261 queries on Stack Overflow in last 12 years.
- Limited community exposure.
- Means of support:
  - Mailing list.
  - Stack Overflow.



[10]

Drawbacks

- Complicated administration.
  - Installation.
  - Maintenance.
- Complicated to run algorithms on cluster.
  - Dependency with localhost.
  - Only Java API support.
- Slow client connection.
- Uses a lot of bandwidth.

Assessment criteria of Accumulo alternatives:

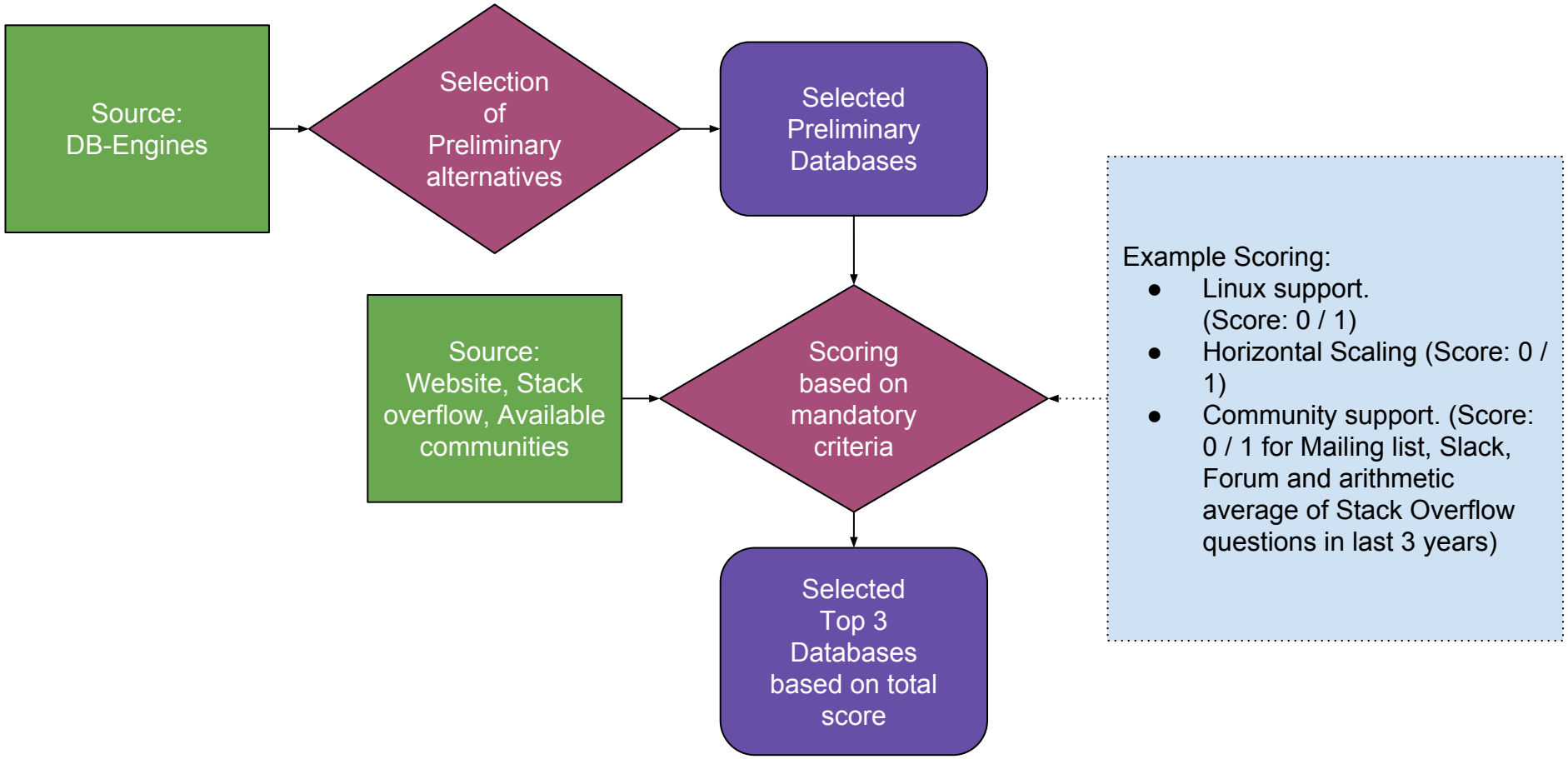
Mandatory Requirements:

- Built-in Linux support.
- Horizontal scaling.
- Fast I/O (current data amount < 1 petabyte).
- Active development and long term support / availability.
- Active Community.
- Java API for client software.

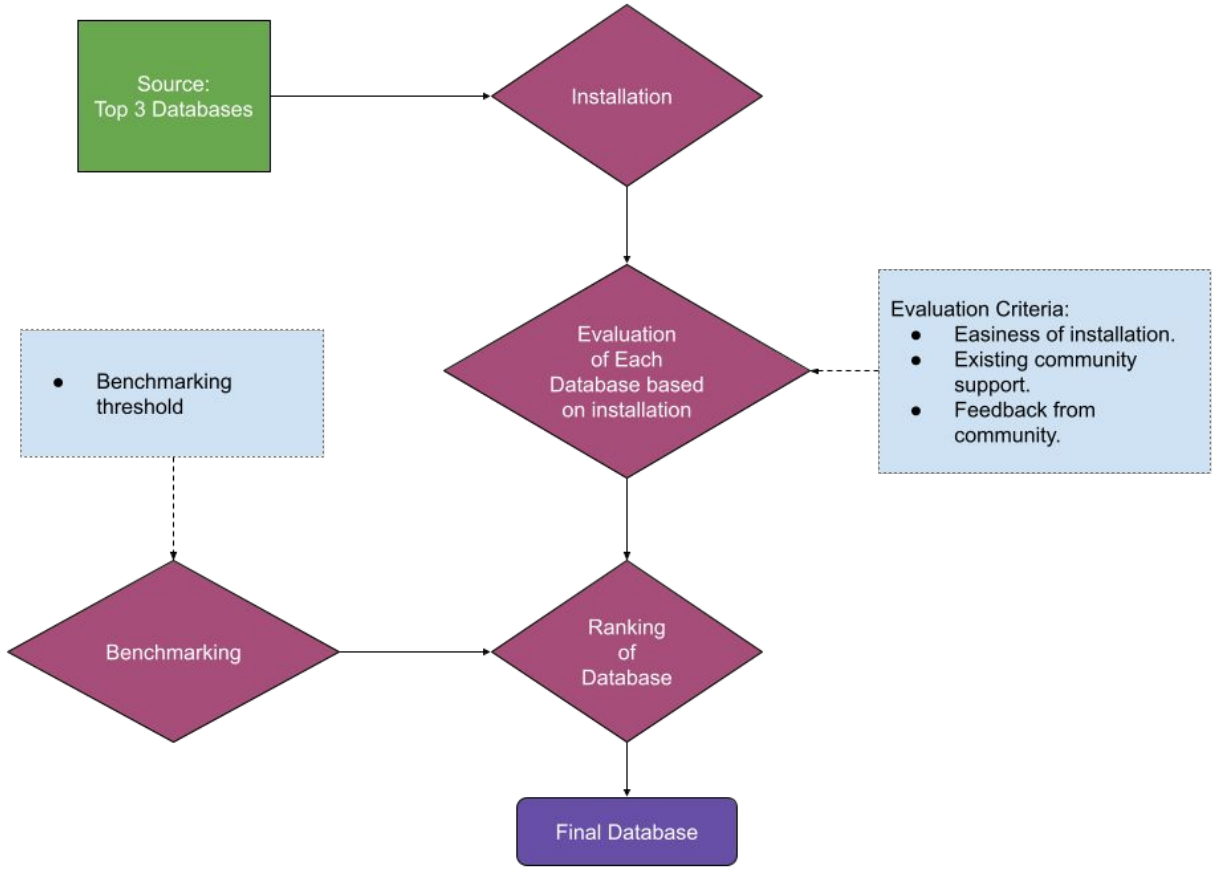
Supplementary Requirements:

- Supported data formats (CSV, Parquet, ...).
- Support for running high throughput analysis algorithms directly on the data.
- Security: Access control and permissions.
- Learning curve.

Research


Source:
DB-Engines
Selection
of
Preliminary
alternatives
Selected
Preliminary
Databases
Source:
Website, Stack
overflow, Available
communities
Scoring
based on
mandatory
criteria
Selected
Top 3
Databases
based on total
score
Example Scoring:
Linux support.
(Score: 0 / 1)
Horizontal Scaling (Score: 0 / 1)
Community support. (Score: 0 / 1 for Mailing list, Slack, Forum and arithmetic average of Stack Overflow questions in last 3 years)

Installation and Benchmarking



Roadmap

	April	May					June				July	
	CW17	CW18	CW19	CW20	CW21	CW22	CW23	CW24	CW25	CW26	CW27	CW28
Preparation for Sprint												
Finding Alternatives												
Installation												
Benchmarking												
Review + make presentation												

References

1. https://accumulo.apache.org
2. https://research.google/pubs/pub27898
3. https://www.geomesa.org
4. https://upload.wikimedia.org/wikipedia/commons/b/ba/Apache_Accumulo_Logo.svg
5. https://upload.wikimedia.org/wikipedia/commons/3/38/Hadoop_logo_new.svg
6. https://upload.wikimedia.org/wikipedia/commons/7/77/Apache_ZooKeeper_logo.svg
7. https://incubator.apache.org/projects/accumulo.html
8. https://en.wikipedia.org/wiki/Apache_Accumulo
9. https://github.com/apache/accumulo
10. https://stackoverflow.com/tags

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Thank You

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